In-class Assignment L6-1 (Group of 3 or less): Hopcroft-Karp Algorithm

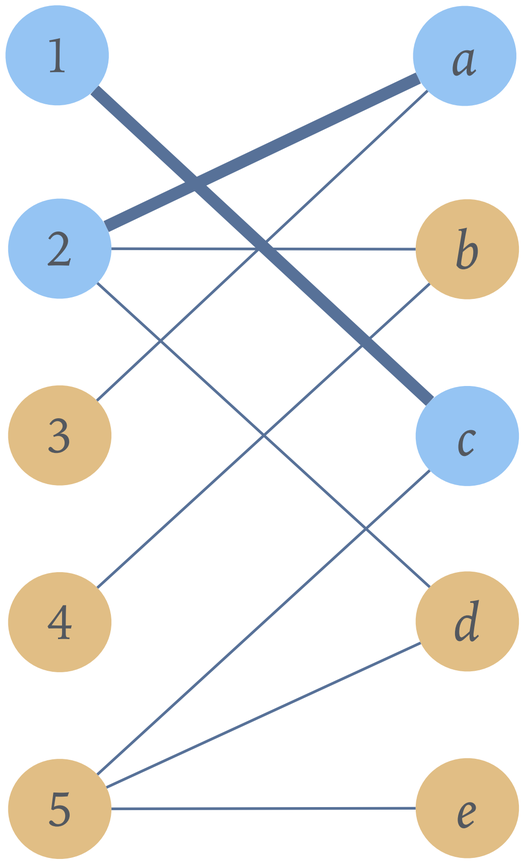
(25 minutes) Given a current matching $M = \{(1,c),(2,a)\}$ on the given graph G, work out Hopcroft-Karp algorithm to determine the maximum matching in G.

In each iteration of lines 3 and 4,

- Draw the digraph G_M
- Draw in DAG H obtained from running BFS on G_M
- Identify a maximal set of vertex-disjoint shortest M-augmenting paths from DAG H
- Show how the current matching M is improved in G.

1st iteration

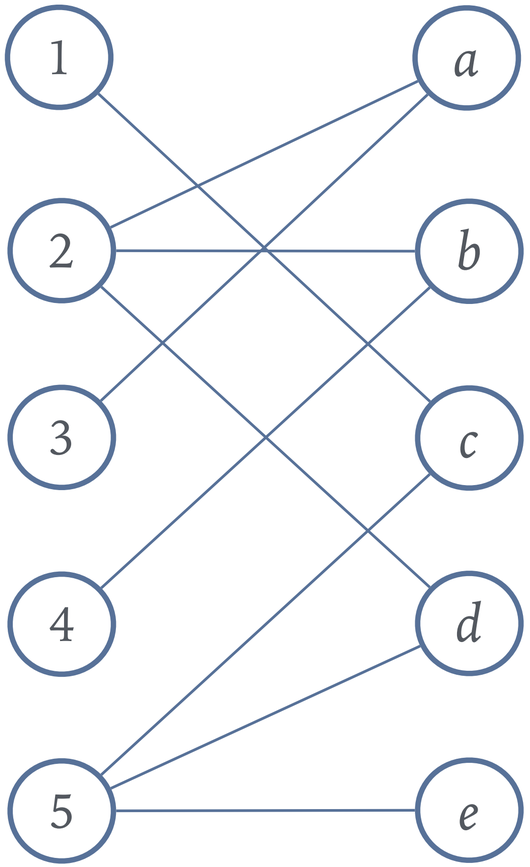
Digraph G_M



DAG $H = (V_H, E_H)$

Maximal set of vertex-disjoint shortest M-augmenting paths

2nd iteration



DAG $H = (V_H, E_H)$

Maximal set of vertex-disjoint shortest M-augmenting paths

3rd iteration

